

Multi-Sensor Environmental Dashboard (ESP32 + Blynk)

An IoT-based environmental monitoring system built with an ESP32 microcontroller. Four sensors — DHT22, LDR, PIR, and HC-SR04 — track temperature, humidity, light intensity, motion, and proximity in real time, with all data displayed on a live Blynk IoT dashboard. An event-driven alert system sends push notifications when motion is detected or an object gets too close. The project is developed and simulated entirely in the Wokwi simulator, running inside VS Code with PlatformIO, so no physical hardware is required.

How It Works

Sensors (DHT22, LDR, PIR, HC-SR04) → ESP32 → WiFi → Blynk Cloud → Mobile app / Web dashboard + Push notifications

- 1 The ESP32 reads all four sensors every 2 seconds.
- 2 Temperature, humidity, light level, motion status, and distance are sent to the Blynk Cloud.
- 3 The Blynk dashboard displays all values live on gauges, an LED indicator, and a history chart.
- 4 When motion is newly detected, or an object comes closer than the set threshold, the ESP32 triggers a Blynk event, which sends a push notification to the phone.

Components Used

Component	Purpose
ESP32	Main controller with built-in WiFi
DHT22	Measures temperature and humidity
LDR (Photoresistor)	Measures ambient light level
PIR Motion Sensor	Detects movement
HC-SR04	Measures distance (ultrasonic)
Blynk IoT	Mobile app and cloud dashboard, with push notification alerts
Wokwi	Online simulator (replaces real hardware)
PlatformIO (VS Code)	Development environment used to build and run the simulation

Circuit Connections

DHT22 (Temperature + Humidity)

DHT22 Pin	Connect To
VCC	ESP32 3V3
SDA	ESP32 GPIO 15
GND	ESP32 GND

LDR (Light Sensor)

LDR Pin	Connect To
VCC	ESP32 3V3

LDR Pin	Connect To
AOUT	ESP32 GPIO 34
GND	ESP32 GND

PIR (Motion Sensor)

PIR Pin	Connect To
VCC	ESP32 5V
OUT	ESP32 GPIO 27
GND	ESP32 GND

HC-SR04 (Ultrasonic/Distance Sensor)

HC-SR04 Pin	Connect To
VCC	ESP32 5V
TRIG	ESP32 GPIO 18
ECHO	ESP32 GPIO 19
GND	ESP32 GND

Combined Pin Summary

Sensor	Signal Pin	ESP32 Pin	Power	Ground
DHT22	SDA	GPIO 15	3V3	GND
LDR	AOUT	GPIO 34	3V3	GND
PIR	OUT	GPIO 27	5V	GND
HC-SR04	TRIG	GPIO 18	5V	GND
HC-SR04	ECHO	GPIO 19	5V	GND

Blynk Setup

- 1 Create a template named **Multi-Sensor Dashboard** (Hardware: ESP32, Connection: WiFi).
- 2 Add five datastreams: **Temperature** — V0, Double; **Humidity** — V1, Double; **Light Level** — V2, Integer; **Motion Status** — V3, Integer; **Distance** — V4, Double.
- 3 Add two Events (for push notification alerts): **motion_alert** — "Motion Detected"; **object_close** — "Object Too Close".
- 4 Add widgets to the dashboard: Gauge for Temperature (V0), Humidity (V1), Light Level (V2), Distance (V4); LED widget for Motion Status (V3); Chart widget linking the datastreams, for historical trends.
- 5 Create a device from the template and copy the Template ID, Template Name, and Auth Token.

How to Run

- 1 Open the project folder in VS Code with the PlatformIO and Wokwi Simulator extensions installed.
- 2 Open **platformio.ini** and confirm the required libraries are listed (Blynk, DHT sensor library).
- 3 Open **src/main.cpp** and replace the three placeholder Blynk values with your own.

- 4 Build the project (PlatformIO: Build).
- 5 Run the simulation (Command Palette → **Wokwi: Start Simulator**).
- 6 In Wokwi, trigger the PIR sensor and move the HC-SR04's distance slider closer than 10 cm to test the alerts.
- 7 Watch the Serial Monitor for live sensor readings and alert messages, and check the Blynk dashboard and phone for push notifications.

Code Overview

Part	What It Does
<code>readDistanceCM()</code>	Sends a trigger pulse and measures the echo return time to calculate distance
<code>digitalRead(PIR_PIN)</code>	Reads the PIR sensor's motion status
<code>Blynk.logEvent(...)</code>	Sends an event to Blynk, which triggers a push notification
<code>lastMotionState</code> / <code>distanceAlertSent</code>	Prevent the same alert from being sent repeatedly while the condition stays true
<code>timer.setInterval(2000L, readAllSensors)</code>	Repeats the full sensor read-and-send cycle every 2 seconds

Tools and Technologies

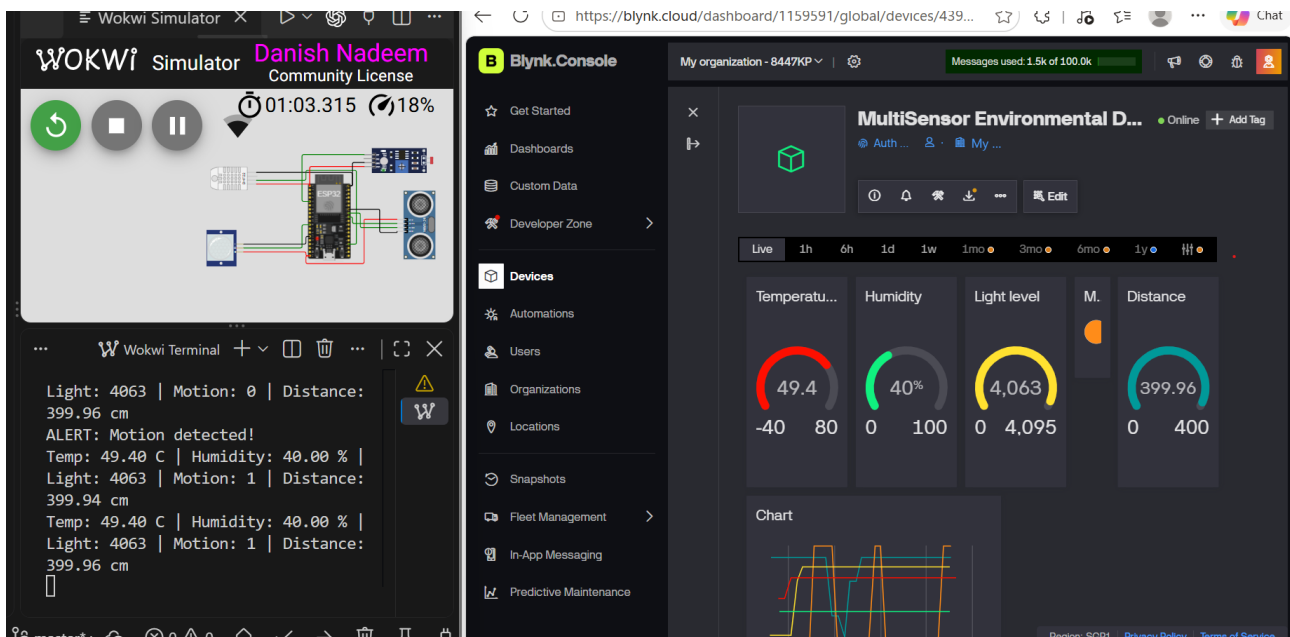
ESP32, Arduino C/C++, Blynk IoT, Wokwi Simulator, PlatformIO, VS Code

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Working Screenshot



Wokwi simulation (left) with live serial output, and the Blynk dashboard (right) showing temperature, humidity, light level, motion status, distance, and the history chart.